

ADVANCED MACHINE LEARNING · WORKSHEET 13

Time Series Analysis

Stationarity · Differencing · AR, ACF/PACF and MA

STUDENT COPY — Lecture 13

Name:

Roll No.:

Date:

PART I: WHY TIME MATTERS

In every model we have built so far — from Linear Regression to Regularization — we assumed that the order of rows in our dataset does not matter. Time Series Analysis begins where that assumption breaks down.

When the order of data matters

▷ HOOK

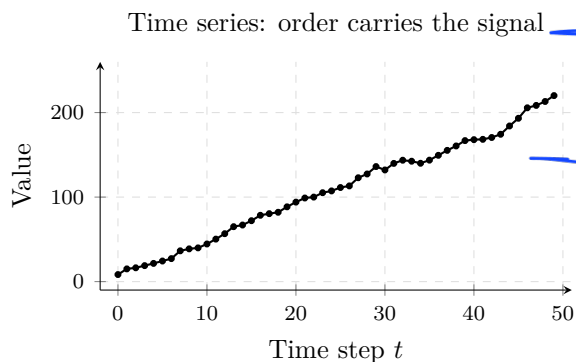
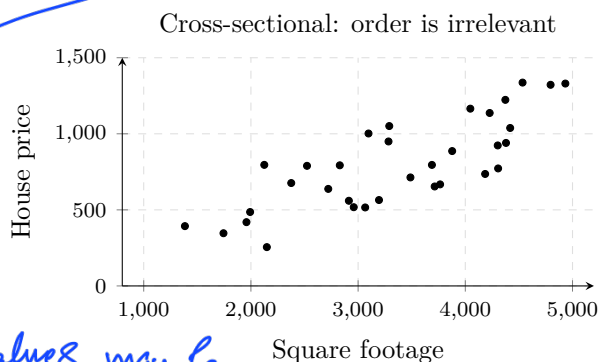
Consider two datasets. The first contains 500 house listings with features like area, bedrooms, and location. You randomly shuffle the rows before training a Linear Regression model. The predictions remain identical. Now consider a second dataset: daily server traffic on your university network recorded every hour for a year. You shuffle the rows. What happens to the data?

Independent observations vs. temporal sequences

In classical Machine Learning, we assume observations are **Independent and Identically Distributed (I.I.D.)**. The order of rows does not matter. Row 1 (a house in Delhi) is completely independent of Row 2 (a house in Mumbai). Each data point is an isolated, standalone event.

Classroom question: If you take a dataset of daily stock prices and completely randomize the order of the rows, what happens to the data?

For temporal data, the order of the rows is not an arbitrary index — it is a vital piece of information. The fact that row 5 happened immediately after row 4 is the most important feature of the dataset.



The values may be dependent on previous time stamp value. } I can't shuffle the data, otherwise Data leakage.

Shuffling the rows on the left changes nothing. Shuffling the rows on the right destroys the information the model needs.

★ KEY INSIGHT

Time Series Analysis is a branch of Machine Learning where time is the primary dimension and the past directly influences the future. Shuffling the data destroys its meaning — unlike cross-sectional data where row order is irrelevant.

What is a time series?

A **Time Series** is a sequence of data points recorded at successive, **equally spaced** intervals in time.

Key structural differences from cross-sectional data:

Aspect	Cross-sectional data	Time Series data
Row index	Arbitrary ID	Timestamp
Row independence	Independent (I.I.D.)	Dependent (current row depends on previous rows)
Standard K-Fold CV	Valid	Invalid (causes data leakage)

Why can we not use standard K-Fold cross-validation for time series?

If you randomly split a time series into training and testing sets, you might end up training on data from 2026 to predict data from 2024. This violates the **arrow of time** and creates **data leakage**.

◆ TAKEAWAY

Time series data replaces the concept of an arbitrary row ID with a specific timestamp. Because the current row depends on previous rows, standard cross-validation techniques cannot be applied directly. Shuffling temporal data destroys its structure.

Components of a time series

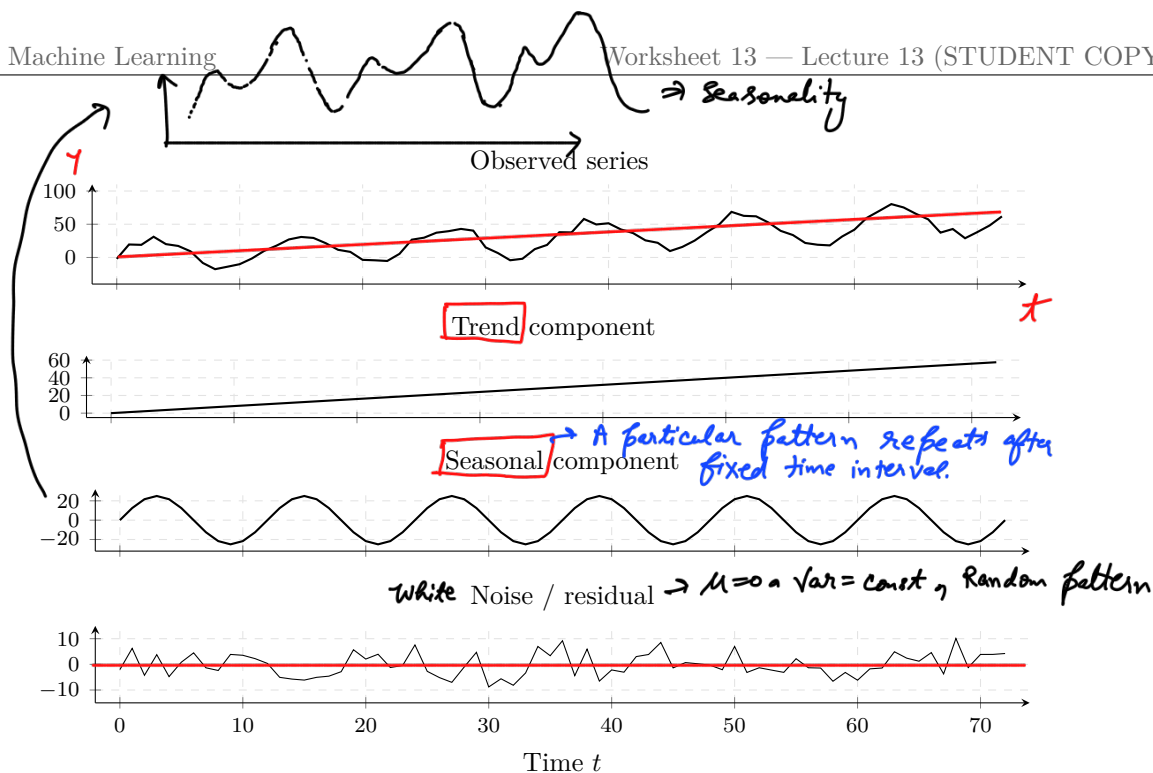
▷ HOOK

Imagine plotting the electricity consumption of your university campus over the past five years. You would likely see usage drop on weekends, drop during summer holidays, peak during exam weeks, and yet show an overall upward trend as more computers and servers are added each year. The observed line on the graph is not one chaotic pattern — it is actually several distinct patterns stacked on top of each other.

Decomposing a time series

An observed time series can be mathematically decomposed into three main components:

- 1. Trend:** The overall, long-term direction of the data. Ignoring day-to-day bumps, is the general trajectory moving **upward** or **downward**?
- 2. Seasonality:** Short-term, repeating cycles tied to specific timeframes. These repeat at known, **fixed** frequencies.
- 3. Noise (Residual / Random Shocks):** The **random**, unpredictable variations that cannot be explained by trend or seasonality.



Classroom question: If a model's prediction errors (residuals) look exactly like pure random noise — no trend, no repeating pattern — what does that tell us about the model?

★ KEY INSIGHT

When we build forecasting models, the algorithm tries to extract the rules of the trend and the cycles of the seasonality. If the model's residuals are pure random noise, the model has captured all systematic information in the data.

◆ TAKEAWAY

Every time series can be decomposed into Trend (long-term direction), Seasonality (repeating cycles), and Noise (random, unpredictable fluctuations). The goal of a forecasting model is to capture trend and seasonality so that only noise remains in the residuals.

PRACTICE P1 · TIME SERIES FUNDAMENTALS

(a) **Identify the component.** For each observation, classify it as Trend, Seasonality, or Noise.

Observation	Component
Sales of umbrellas spike every monsoon season.	<u>S</u>
A country's GDP has grown steadily over the past 30 years.	<u>T</u>
An unexpected factory fire disrupts production for one week.	<u>N</u>
Electricity demand peaks every afternoon and dips every night.	<u>S</u>
Global average temperature has risen over the past century.	<u>T</u>
A sudden viral social media post causes a one-day sales spike.	<u>N</u>

(b) **True or False.** Justify each answer.

Statement	T / F	Justification
Shuffling the rows of a time series dataset does not affect model predictions.	☐ F	_____
Classical time series models generally assume equally spaced time intervals.	☐ T	_____
Standard K-Fold cross-validation can be used for time series without modification.	☐ F	_____
If a model's residuals show a repeating pattern, the model has captured all seasonality.	☐ F	_____

(c) **Short reasoning.** Why is a house price dataset considered cross-sectional but a daily stock price dataset considered a time series? Explain in terms of the I.I.D. assumption.

PART II: MAKING DATA READY FOR FORECASTING

Classical time series models require the data to have stable statistical properties. This section introduces stationarity and the techniques to achieve it.

Stationarity

▷ HOOK

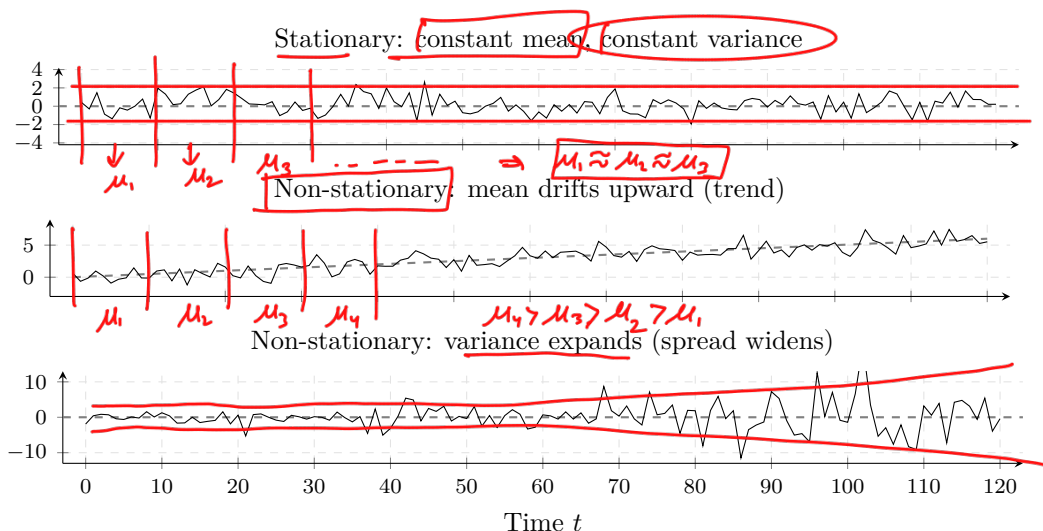
A tech startup's revenue has been growing by exactly 20% every year for the past decade. You want to predict next year's revenue. Can you simply take the arithmetic mean of the last ten years of revenue? The average would be far lower than the current reality because the baseline is constantly moving upward. For our forecasting models to work, the statistical properties of the data must remain stable over time.

Stationary vs. non-stationary series

A **Stationary Time Series** is one whose statistical properties do not change over time. Specifically:

Property	Requirement	In plain language
<u>Mean</u>	Constant	Data fluctuates around a flat, horizontal line
<u>Variance</u>	Constant	The spread of fluctuations stays the same over time
<u>Autocovariance</u>	Constant	The relationship between observations depends on <u>the lag</u> , not on the particular point in time

A **Non-Stationary Time Series** is one where the rules are changing. The mean is **drifting** (trend), or the variance is **expanding** (like a megaphone shape on a graph).



Classroom question: Why do Machine Learning algorithms struggle with non-stationary data?

★ KEY INSIGHT

Machine Learning algorithms are pattern-matching engines. If the rules governing the data are constantly changing (non-stationary), the algorithm cannot reliably learn from the past to predict the future. Stationarity ensures that patterns learned from historical data remain valid for future predictions.

◆ TAKEAWAY

Classical time series models require stationary data — data with constant mean, constant variance, and constant autocovariance. Most real-world data is non-stationary, so it must be transformed before modeling.

Lags: the features of time series

Non-Stationary $\rightarrow$ Stationary
(Lag Differencing)

▷ HOOK

In standard regression, we predict house price using features like area, bedrooms, and location. But in time series, we have only one column: the past values of our target variable. If you want to predict today's maximum temperature and are allowed only one piece of historical data, what would you ask for? Most likely, yesterday's temperature. This past observation — called a lag — becomes our feature.

t	y_t	Lag1 (y_{t-1})	Lag2 (y_{t-2})	Lag3 (y_{t-3})	Δy_t
1	10				
→ 2	20	10			10
3	30	20	10		10
$K+1=4 \rightarrow$ 4	40	30	20	10	10
5	50	40	30	20	10
6	60	50	40	30	10
7	70	60	50	40	10

$$\Delta y_t = y_t - y_{t-1}$$

$\hookrightarrow$ Differenced Series

$\downarrow$
We can try to achieve stationarity.

Defining lags

Notation	Name	Meaning
y_t	Current observation	Today's value
y_{t-1}	Lag 1	Yesterday's value
y_{t-2}	Lag 2	Day before yesterday's value
y_{t-7}	Lag 7	Same day last week (weekly seasonality)
y_{t-k}	Lag k	Value k time steps ago

By defining lags, we solve a major engineering problem: we represent the past mathematically as distinct variables. This allows us to feed historical information into an equation just like we feed features into a regression model.

◆ TAKEAWAY

Lags are the features of time series. They convert the temporal structure into a format that regression-style models can use. Lag 1 is the most recent past observation; Lag k is the observation k steps ago.

Making a series stationary: Differencing

▷ HOOK

We have a contradiction: our models require stationary data, but real-world data typically has strong trends and seasonality. How do we bridge this gap? Instead of predicting the absolute value, we predict the change from one time step to the next.

Lag differencing

If the data has a trend (constantly moving), we transform it by subtracting the previous observation from the current one. This technique is called **Lag Differencing**.

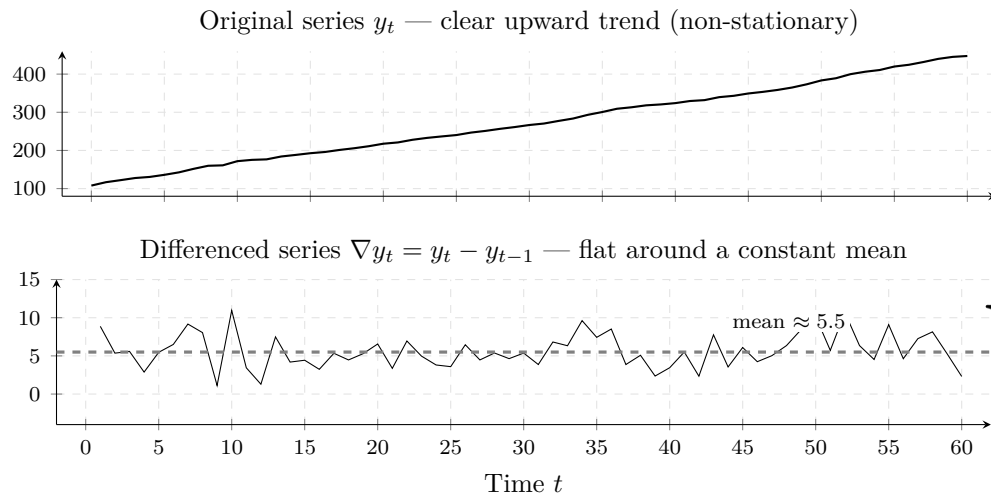
$$\Delta Y_t = Y_t - Y_{t-1}$$

Symbol	Meaning
ΔY_t	The new, differenced value at time t
Y_t	The original, raw observation at time t
Y_{t-1}	The original observation at the previous time step

Numerical example

Time (t)	Raw Stock Price y_t	Differenced Series ΔY_t
Day 1	100	— (no prior day)
Day 2	105	105 − 100 = 5
Day 3	110	110 − 105 = 5
Day 4	115	115 − 110 = 5
Day 5	122	122 − 115 = 7

The original data (100, 105, 110, 115, 122) has a clear **upward** trend. The differenced data fluctuates around a roughly constant mean, making it **stationary**.



★ KEY INSIGHT

Differencing removes the trend by converting absolute values into period-to-period changes. The resulting series fluctuates around a constant mean, satisfying the stationarity requirement.

◆ TAKEAWAY

Lag differencing ($\Delta Y_t = Y_t - Y_{t-1}$) transforms non-stationary data into stationary data by removing the trend. Instead of modeling the raw values, we model the changes between consecutive time steps.

PRACTICE P2 · STATIONARITY, DIFFERENCING, AND LAGS

(a) **Diagnose stationarity.** For each description, state whether the series is stationary or non-stationary and identify the violated property.

Description	Stationary?	Which property is violated?
Data fluctuates randomly around a constant value of 50 with similar spread throughout.	<u>Yes</u>	_____
Sales show a steady upward climb over five years.	<u>No</u>	_____
Stock returns oscillate around zero, but the swings become wider each year.	<u>No</u>	_____
Hourly temperature readings show no trend and similar variation all year.	<u>Yes</u>	_____

(b) **Apply differencing.** Complete the table.



Month	Monthly Sales y_t	Differenced $\Delta Y_t = Y_t - Y_{t-1}$
Jan	200	—
Feb	220	20
Mar	250	30
Apr	245	-5
May	270	25
Jun	300	30

(i) Does the original series appear stationary? No Why?

Trend

(ii) Does the differenced series appear more stationary? _____ Why?

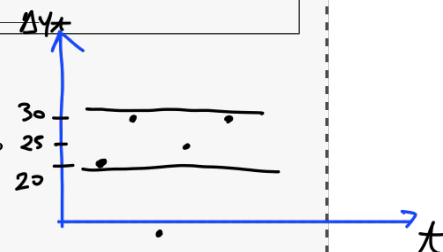
(c) **Identify the lag.** Express each statement using lag notation.

Statement	Lag notation
"The temperature two days ago"	<u>Lag 2</u>
"Last week's sales (weekly data)"	<u>Lag 7</u>
"The observation three time steps ago"	<u>Lag 3</u>

(d) **Assertion–Reason.**

	Statement	True / False
Assertion	Differencing is used to make a non-stationary series stationary.	☐ <u>T</u>
Reason	Differencing removes the trend by converting absolute values into period-to-period changes.	☐ <u>T</u>

The Reason is a correct explanation of the Assertion: ☐



PART III: BUILDING FORECASTING MODELS

With stationary data and lag features defined, we can now build models that predict the future using the past.

Auto Regression (AR) models → Build on stationary data

▷ HOOK

Recall Multiple Linear Regression: we predicted a target Y using feature columns X_1, X_2, X_3 . But in time series, we do not have separate feature columns like "area" or "bedrooms." We have only one column — the past values of our target variable. What if we treat our lags exactly like the features in a regression model? We would be running a regression of a variable against historical versions of itself.

The AR model equation

The **Auto Regression (AR)** model predicts the current value as a weighted sum of its own past values. The prefix "Auto" means self. We are running a regression of a variable against historical versions of itself.

The Auto Regression equation of order p , denoted $\boxed{\text{AR}(p)}$:

$$\boxed{y_t = c + \phi_1 \cdot \boxed{y_{t-1}} + \phi_2 \cdot \boxed{y_{t-2}} + \cdots + \phi_p \cdot \boxed{y_{t-p}} + \varepsilon_t}$$

$p \rightarrow$ order of AR Model

Symbol	Meaning
y_t	The current value we are trying to predict today
c	A constant intercept (the baseline mean of the series)
y_{t-1}	Lag 1 — yesterday's value, acting as Feature 1
ϕ_1	The coefficient (weight) assigned to Lag 1
p	The order of the model — how many lags we look back
ε_t	White noise — the unpredictable error term for today

Example: An AR(2) model for daily temperature prediction would be:

$$\boxed{y_t = c + \phi_1 \cdot y_{t-1} + \phi_2 \cdot y_{t-2} + \varepsilon_t}$$

This uses **two** lags (yesterday and the day before yesterday) as features.

★ KEY INSIGHT

The AR model operates on a principle of **momentum**: if the data had high values yesterday and the day before, the positive weights (ϕ) carry that momentum forward to predict a high value for today. It transforms the problem of predicting over time into a standard weighted sum — like Linear Regression where the features are the variable's own past values.

The problem of choosing p

Classroom question: What happens if we set $p = 365$, forcing the model to learn 365 weights for predicting tomorrow's temperature?

Setting p too **large** leads to overfitting. Setting p too **small** may miss important patterns. We need a mathematical tool to determine which past lags carry statistically significant signals.

◆ TAKEAWAY

The $\text{AR}(p)$ model predicts the current value as a linear combination of p past values plus an error term. It converts time series forecasting into a regression problem where lags serve as features. The critical question is choosing the right number of lags p .

Autocorrelation (ACF) and Partial Autocorrelation (PACF)

▷ HOOK

We need to select p — the number of lags for our AR model — but we cannot just guess. We need to objectively measure the strength of the relationship between observations separated by different time gaps. You already know Pearson correlation, which measures the relationship between two different variables. What if we apply the same idea, but between a variable and a lagged version of itself?

Autocorrelation Function (ACF)

Autocorrelation measures the linear relationship between a variable and a **lagged version** of itself.

The ACF at lag k calculates the correlation between the original series y_t and the shifted series y_{t-k} .

The formula for ACF at lag k :

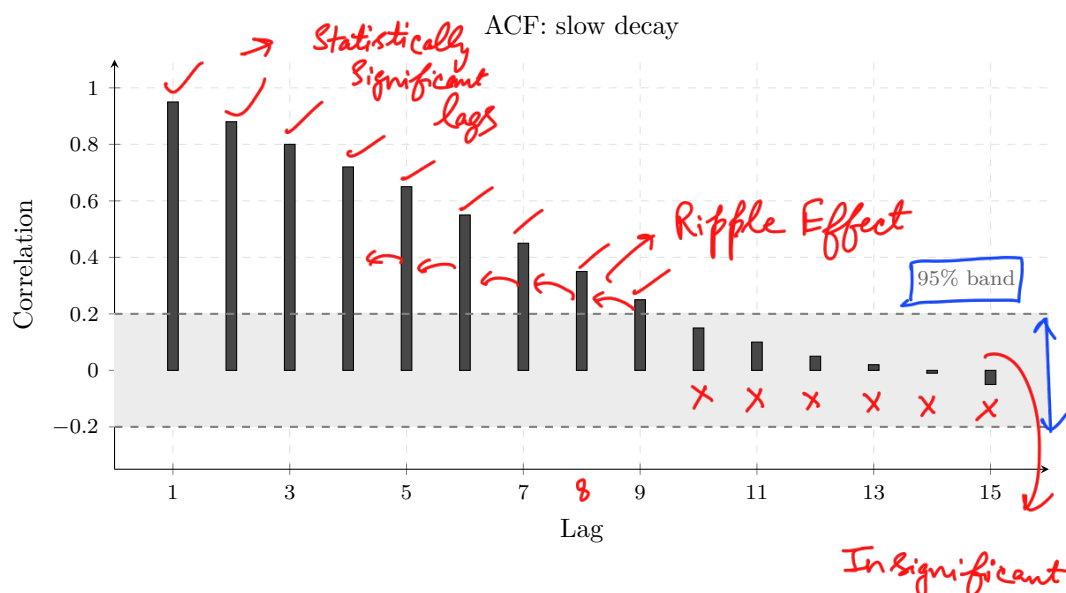
$$\rho_k = \frac{\sum_{t=k+1}^N (Y_t - \bar{Y})(Y_{t-k} - \bar{Y})}{\sum_{t=1}^N (Y_t - \bar{Y})^2}$$

$$\rho = \frac{\text{Cov}(X, Y)}{\sigma_x \cdot \sigma_y}$$

$$\rho = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

where $\bar{Y}$ is the mean of the series and N is the total number of observations.

ACF value	Interpretation
Near +1 <u> </u>	Strong positive correlation: when the past went up, today tends to go up (trend momentum)
Near -1 <u> </u>	Strong negative correlation: when the past went up, today tends to go down (mean-reverting behavior)
Near 0 <u> </u>	No linear relationship: that specific past lag offers no useful information for forecasting today



ACF involves the direct as well as indirect effect of previous lags.

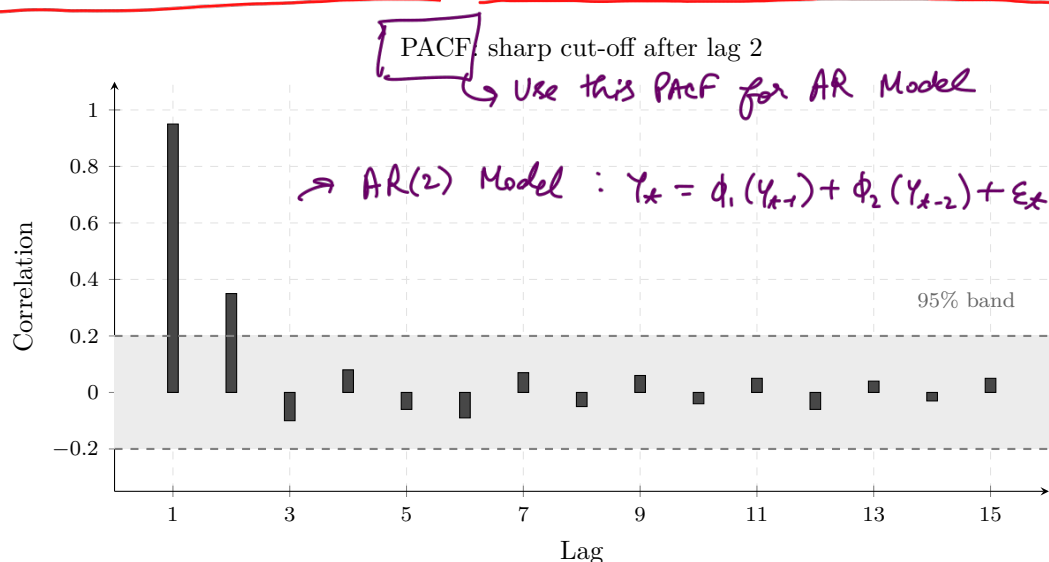
The problem with ACF: indirect correlations

Classroom question: Today's temperature is highly correlated with yesterday's. Yesterday's was highly correlated with the day before. Standard ACF will therefore show that today is correlated with the day before yesterday. Is that a direct relationship, or a ripple effect passing through yesterday?

Standard ACF captures the **total** effect (both direct and indirect). It might tell us we need 10 lags when in reality we only need 1 lag carrying a chain reaction forward.

Partial Autocorrelation Function (PACF)

PACF measures the pure, **direct** correlation between y_t and y_{t-k} after completely removing the influence of all **intermediate** lags in between. It isolates the unique new information that lag k brings.



ACF decays slowly — lag 1 correlation ripples through to lag 2, lag 3 and beyond. PACF strips those ripples out and cuts off sharply after lag 2, revealing only the direct dependencies.

How to read a PACF plot to choose p

1. Look at the PACF plot.
2. Identify the shaded confidence band (the significance threshold).
3. Count how many lag bars extend significantly **beyond** the band before they drop into the noise.
4. If only Lag 1 and Lag 2 are significant, set $p = 2$. Build an AR(2) model.

★ KEY INSIGHT

ACF shows the total correlation (direct + indirect) between a series and its lags. PACF removes indirect effects and shows only the direct correlation. To choose p for an AR model, count the number of significant bars in the PACF plot.

Tool	What it measures	Use for
ACF	Total correlation (direct + indirect)	Determining the order q of an MA model (covered next)
PACF	Direct correlation only	Determining the order p of an AR model

fill feedback

◆ TAKEAWAY

ACF measures the total linear relationship between a series and its lags, but includes indirect ripple effects. PACF isolates the direct effect of each lag by removing intermediate influences. Use the PACF plot to determine the order p of an AR model.

PRACTICE P3 · AUTO REGRESSION, ACF, AND PACF

(a) **Write the AR equation.** Write out the full AR(3) equation for predicting daily sales, labeling each term.

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \phi_3 y_{t-3} + \varepsilon_t + c$$

(b) **Interpret a PACF plot.** A PACF plot for a temperature dataset shows the following:

Lag	1	2	3	4	5	6	7
PACF value	0.85	0.42	0.05	-0.02	0.03	-0.01	0.38
Significant?	Yes	Yes	No	No	No	No	Yes

(i) Based on lags 1–6 only, what order AR model would you initially choose? $p = 2$

(ii) Lag 7 is also significant. What real-world pattern might this represent?

(iii) Would you include lag 7 in your model? Why or why not?

(c) **Compare ACF and PACF.** Fill in the blanks.

Property	ACF	PACF
Measures _____ correlation	Total (direct + indirect)	_____
Behavior for an AR process	_____	_____
Used to determine	_____	Order p of the AR model

(d) **MCQ.** An AR(1) model with $\phi_1 = 0.9$ and $c = 2$ predicts y_t . If $y_{t-1} = 50$, what is the predicted value (ignoring the error term)?

(A) 47 (B) 45 (C) 50 (D) 52

Answer: _____

(e) **Error identification.** A student writes: "To choose p for an AR model, I look at the ACF plot and count the significant bars." Identify and correct the error.

PART IV: MODELING EXTERNAL SHOCKS

The AR model assumes the future is driven entirely by the momentum of past values. But real-world systems are constantly hit by unpredictable external events.

Random shocks and Moving Average (MA) models

Stationary Data

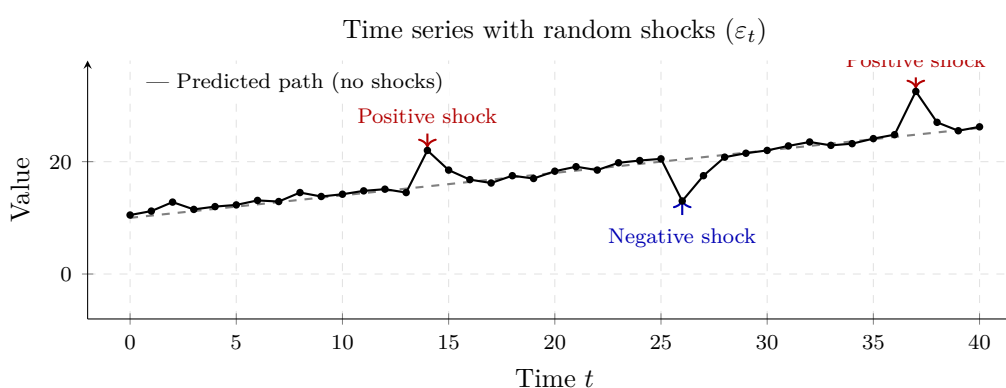
▷ HOOK

Your AR model is predicting daily stock prices based on yesterday's and the day before's values. At 2:00 PM today, a sudden government policy announcement causes a massive market crash. Did yesterday's stock price predict that policy announcement? No — it was completely unpredictable from the historical data. These external, unpredictable events constantly push time series off their natural paths.

Random shocks (innovations)

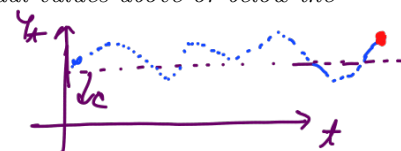
In Time Series Analysis, unpredictable external events are called **Random Shocks** (or **innovations**). Mathematically, random shocks are the **forecast errors** at each time step:

$$\text{Random Shock} = \text{Actual value} - \text{Predicted value} = \varepsilon_t$$



The dashed gray line shows the predicted path. Random shocks (ε_t) push the actual values above or below the prediction at unpredictable times.

This is the ε_t term that appears in the AR equation.



The Moving Average (MA) model

Build on Noise

What if we build a model that learns from the shocks themselves, rather than just the past values?

Important: Do NOT confuse this with a "rolling mean" (like a 50-day moving average on a stock chart). The MA model in forecasting uses **past forecast errors (random shocks)** as its predictors.

The Moving Average equation of order q , denoted $MA(q)$:

$$y_t = c + \varepsilon_t + \theta_1 \cdot \varepsilon_{t-1} + \theta_2 \cdot \varepsilon_{t-2} + \dots + \theta_q \cdot \varepsilon_{t-q}$$

MA(2)

$$y_t = c + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2}$$

Symbol	Meaning
y_t	The value we are trying to predict today
c	The constant mean of the series
ε_t	The current random shock happening right now (unpredictable)
ε_{t-1}	The random shock (forecast error) that happened yesterday
θ_1	The coefficient/weight assigned to yesterday's shock
q	The order of the MA model — how many past shocks we look at

↓
ACF
↑

AR vs. MA: two fundamentally different assumptions

Aspect	AR (Auto Regression)	MA (Moving Average)
Predicts using	Past values $y_{t-1}, y_{t-2}, \dots$	Past errors $\varepsilon_{t-1}, \varepsilon_{t-2}, \dots$
Assumes current values are driven by	SIGNAL of the past	NOISE of the past
Intuition	Momentum: if sales were high yesterday, they will be high today	Shock absorption: a temporary bump slowly fades over q days
Order parameter	p (number of lags)	q (number of past shocks)

★ KEY INSIGHT

AR and MA models capture two fundamentally different dynamics. AR models assume **momentum** — if sales were high yesterday, they will naturally be high today. MA models assume the baseline is mostly flat, but temporary **shocks** (like a good marketing campaign) bump the series off its baseline, and that bump slowly fades over q time steps.

◆ TAKEAWAY

Random shocks are the unpredictable, external events that push a time series off its natural path. The $MA(q)$ model learns from these past forecast errors. AR models use past values (signal); MA models use past errors (noise). In the next lecture, combining both leads to the powerful ARMA and ARIMA models.

PRACTICE P4 · COMPREHENSIVE CHALLENGE (Homework)

(a) Fill in the Key Terms table.

Term	Definition
Time Series	_____
Stationarity	_____
Differencing	_____
Lag	_____
Autocorrelation (ACF)	_____
Partial Autocorrelation (PACF)	_____
$AR(p)$	_____
$MA(q)$	_____
Random Shock	_____

(b) Match the columns.

Concept	Description
(1) Trend	(A) The correlation of a series with a lagged version of itself
(2) Seasonality	(B) An unpredictable external event that causes forecast error
(3) Noise	(C) Short-term repeating cycles at fixed frequencies
(4) Stationarity	(D) A series whose statistical properties do not change over time
(5) ACF	(E) Random, unexplainable variation after trend and seasonality are removed
(6) Random Shock	(F) The long-term direction of the data

Matches: (1)–____, (2)–____, (3)–____, (4)–____, (5)–____, (6)–____

(c) Case study. A retail store records weekly sales data for 3 years. The data shows a steady upward trend and a spike every December (holiday season).

(i) Is this data stationary? _____. Identify the non-stationary components.

(ii) What transformation would you apply first to address the trend?

(iii) After making the data stationary, the PACF plot shows significant bars at lags 1 and 2 only. What model would you build?

(iv) After fitting the model, the residuals show a repeating weekly pattern. What does this indicate?

(d) Compare and contrast. Complete the table.

Aspect	AR model	MA model
Inputs to the model	_____	_____
What it assumes drives current values	_____	_____
Order parameter name	_____	_____
Determined using	_____	_____

(e) Synthesis questions. Answer any three.

(i) Why does shuffling the rows of a time series dataset destroy useful information, but shuffling a cross-sectional dataset does not?

(ii) Explain the relationship between stationarity and the reliability of forecasting models.

(iii) Why is PACF more useful than ACF for selecting the order of an AR model?

(iv) A friend says: "The MA model uses a rolling average of past data to predict the future." Is this correct? Explain.

(v) How does the concept of lags connect time series forecasting to Linear Regression?

(vi) Why is differencing necessary before applying an AR model to data with an upward trend?

BY THE END OF THIS SHEET, YOU CAN

- ☐ Explain why time series data violates the I.I.D. assumption and why row order matters.
- ☐ Decompose an observed time series into Trend, Seasonality, and Noise components.
- ☐ Define stationarity and identify whether a given series is stationary or non-stationary.
- ☐ Apply lag differencing to transform a non-stationary series into a stationary one.
- ☐ Define lags and explain how they serve as features for time series models.
- ☐ Write the $AR(p)$ equation and interpret each term.
- ☐ Distinguish between ACF and PACF and use the PACF plot to select the order p of an AR model.
- ☐ Define random shocks and write the $MA(q)$ equation.
- ☐ Compare AR and MA models in terms of inputs, assumptions, and use cases.

Worksheet 13 · AML · Lecture 13: Time Series Analysis.